

Problem A: Air traffic Control

To improve safety and reduce air traffic controller workload, the Federal Aviation Agency (FAA) is considering adding software to the air traffic control system that would automatically detect potential aircraft flight path conflicts and alert the controller. To that end, an analyst at the FAA has posed the following problems

Requirement A: Given two airplanes flying in space, when should the air traffic controller consider the objects to be too close and to require intervention?

Requirement B: An airspace sector is the section of three-dimensional airspace that one air traffic controller controls. Given any airspace sector, how we measure how complex it is from an air traffic workload perspective? To what extent is complexity determined by the number of aircraft simultaneously passing through that sector (1) at any one instant? (2) during any given interval of time? (3) during a particular time of day? How does the number of potential conflicts arising during those periods affect complexity?

Does the presence of additional software tools to automatically predict conflicts and alert the controller reduce or add to this complexity?

In addition to the guidelines for your report, write a summary (no more than two pages) that the FAA analyst can present to Jane Garvey, the FAA Administrator, to defend your conclusions

问题 A: 空间交通管制

为加强安全并减少空中交通指挥员的工作量，联邦航空局(FAA)考虑对空中交通管制系统添

加软件，以便自动探测飞行器飞行路线可能的冲突，并提醒指挥员。为完成此项工作，**FAA** 的分析员提出了下列问题。

要求 **A**: 对于给定的两架空中飞行的飞机，空中交通指挥员应在什么时候把该目标视为太靠近，并予以干预。

要求 **B**: 空间扇形是指某个空中交通指挥员所控制的三维空间部分。给定任意一个空间扇形，我们怎样从空中交通工作量的方位来估量它是否复杂？当几个飞行器同时通过该扇形时，在下面情形所确定的复杂性会达到什么程度：（1）在任一时刻？（2）在任意给定的时间范围内？（3）在一天的特别时间内？在此期间可能出现的冲突总数是怎样影响着复杂性来的？提出所添加的软件工具对于自动预告冲突并提醒指挥员，这是否会减少或增加此种复杂性？在作出你的报告方案的同时，写出概述(不多于二页)使 **FAA** 分析员能提交给 **FAA** 当局 **Jane Garvey**，并对你的结论进行答辩。